Alternating Iterative Refinement

Design and Analysis of Algorithms
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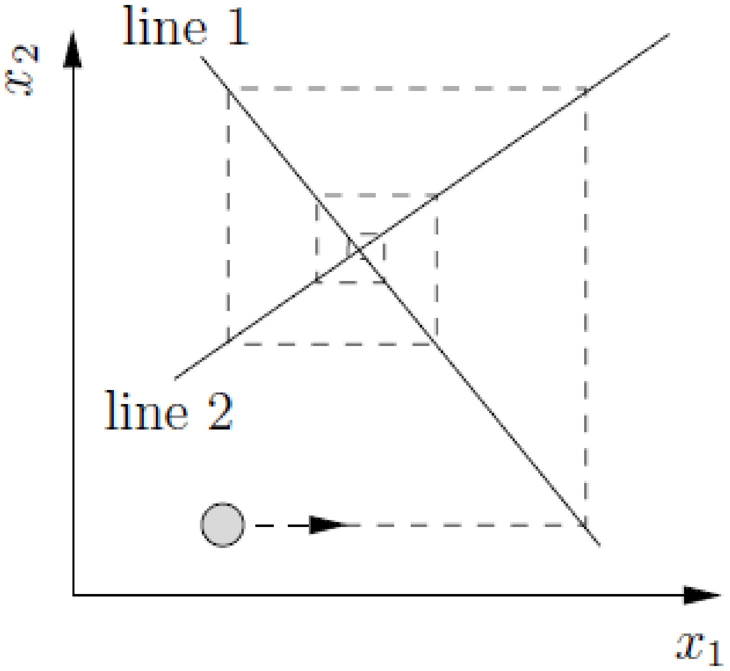
Coordinate Descent

- Minimizing $f(x_1, x_2)$ by:
  - Holding x_1 fixed, minimizing as a function of x_2
  - Holding x_2 fixed, minimizing as a function of x_1
  - Repeat
- **Example:** Gauss-Seidel method for iteratively solving a linear system.
  - Solve equation 1 for an updated value of x_1
  - Solve equation 2 for an updated value of x_2
  - Etc.



Linear Systems

- N unknown variables, N equations
- Example: $x + y = 3$

  $2x - y = 3$
- Example: $x_1 + 2x_2 + 3x_3 = 10$

  $4x_1 + 5x_2 + 6x_3 = 11$

  $7x_1 + 8x_2 + 9x_3 = 12$
- We often write these using matrix-vector notation: $\underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 10 \\ 11 \\ 12 \end{bmatrix}}_{b}$
- $O(N^3)$ to solve with exact methods like Gaussian elimination; slightly faster with divide & conquer.
- For large (typically sparse) system, iterative methods can be a significant win in terms of efficiency.

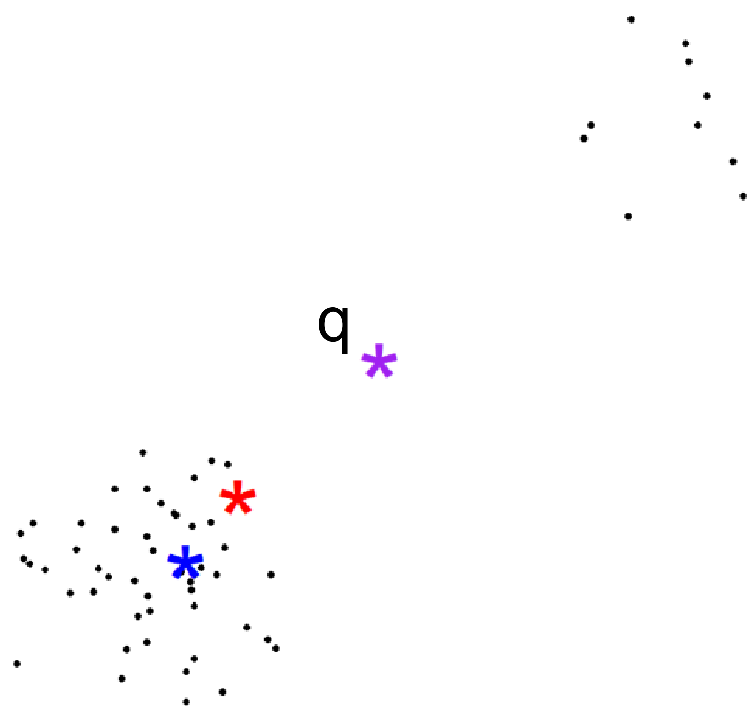
Finding a "Center" of a Point Set

Find center point **q** minimizing:

$\sum_i ||q - x_i||$ (median)

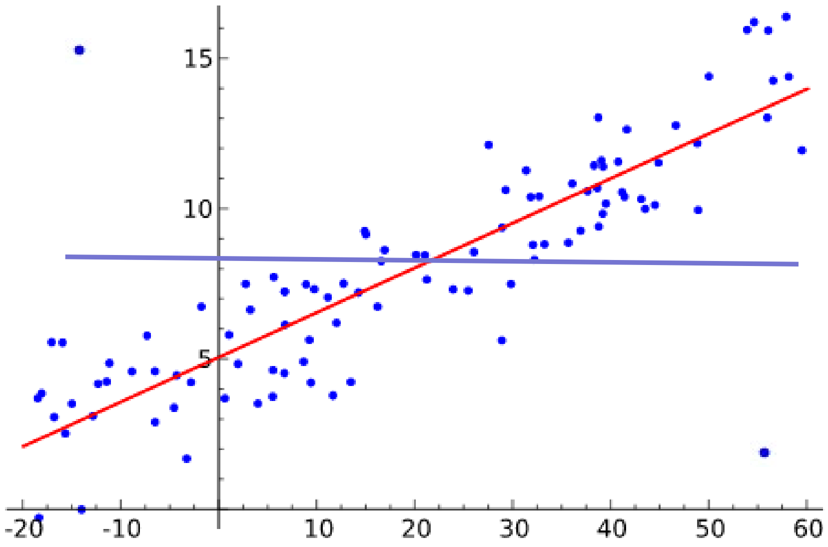
$\sum_i ||q - x_i||^2$ (mean)

$\max_i ||q - x_i||$ (cent

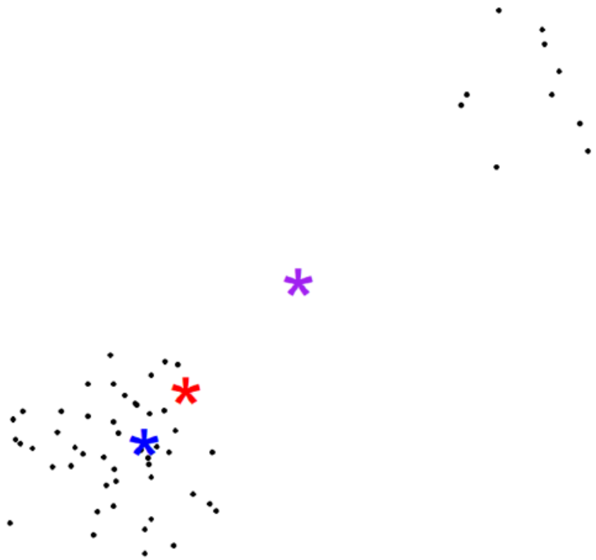


Outlooks on Outliers

Regression



Finding a "Center"



Outliers **BAD**: goal is to fit model being insensitive to outliers

- Outliers represent part of your data that has been corrupted by noise or an adversary

Outliers **GOOD**: goal is to fit a model possibly paying more attention to outliers

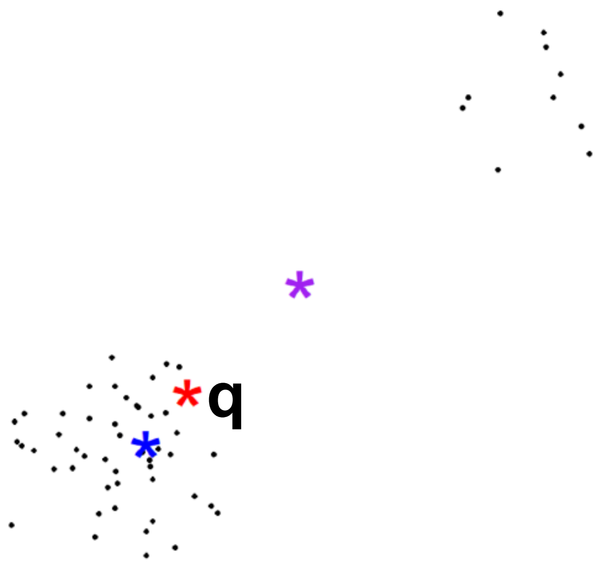
- Outliers represent important albeit rare events that need to be reflected in our model in order for it to be considered sufficiently robust.

Iteratively Reweighted Least Squares

To find center point **q** minimizing:

$f(q) = \sum_i ||q - x_i||^2$

Just set $f'(q) = 0$,
which gives $2 \sum_i (q - x_i) = 0$,
telling us to set $q = \sum_i x_i / n$.

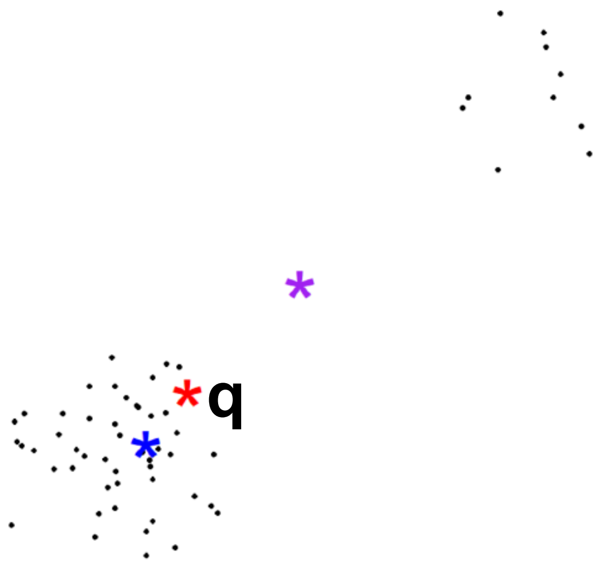


Iteratively Reweighted Least Squares

To find center point **q** minimizing:

$f(q) = \sum_i w_i ||q - x_i||^2$

Just set $f'(q) = 0$,
which gives $2 \sum_i w_i(q - x_i) = 0$,
telling us to set $q = \sum_i w_i x_i / \sum_i w_i$.



Similarly, to find center point **q** minimizing:

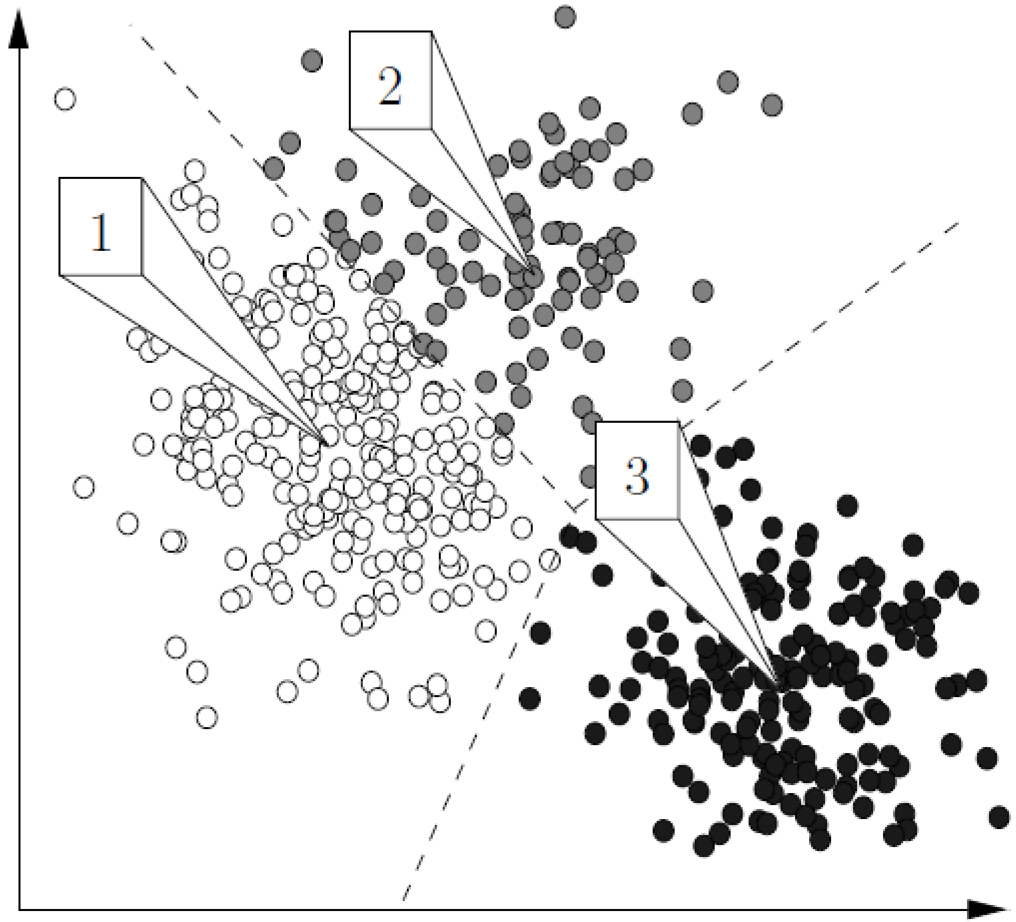
$f(q) = \sum_i ||q - x_i||^p$

Setting $f'(q) = 0$ gives $2 \sum_i p||q - x_i||^{p-2} (q - x_i) = 0$,
telling us we'd have obtained the right answer if we started with a weighted least squares fit, being prescient enough to set $w_i = p||q - x_i||^{p-2}$.

So alternate updating q and the w_i values!

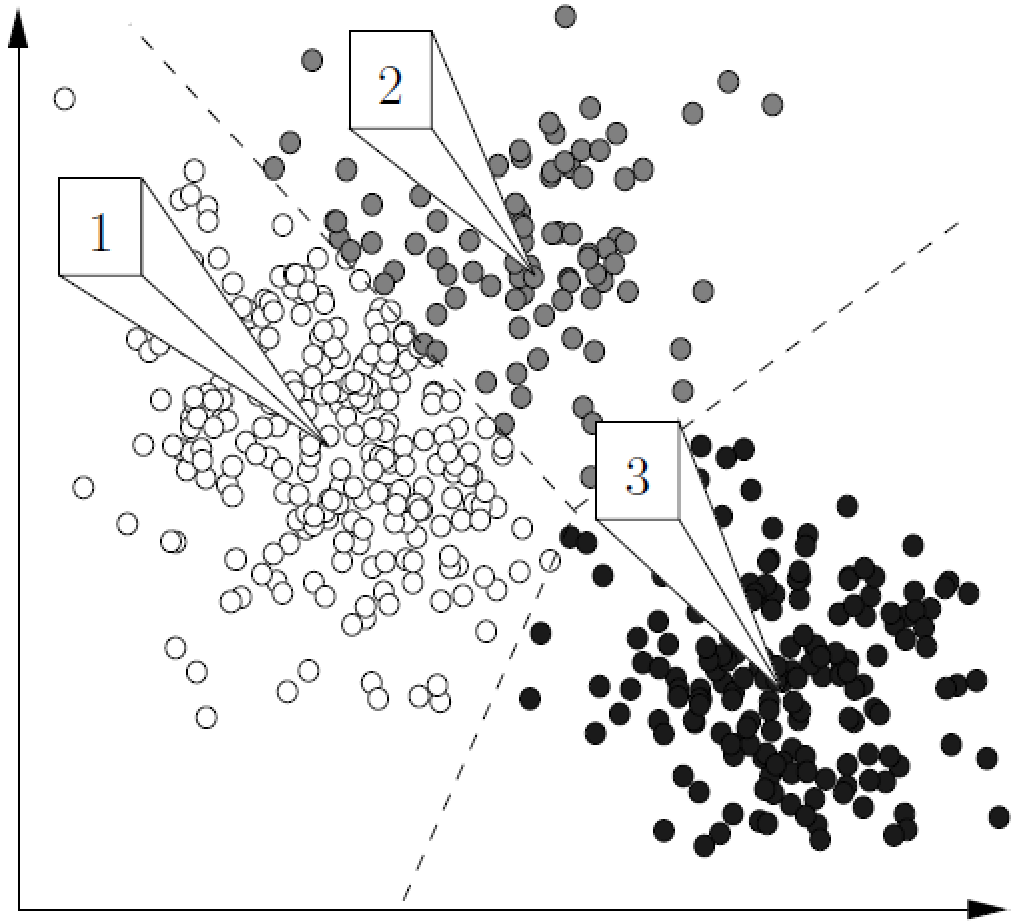
Center-Based Clustering (e.g., k-Means Clustering)

- **Goal:**
  - Partition points into k clusters
  - Determine a center for each cluster; (e.g., minimize sum of squared distances from points to their respective cluster centers)
- Given centers, easy to determine point → cluster assignments
- Given point → cluster assignments, easy to determine centers
- Iteratively update in alternation!



Fuzzy Center-Based Clustering (e.g., Fuzzy k-Means Clustering)

- **Goal:**
  - Fractionally assign points to clusters
  - Determine a center for each cluster
- Given point → cluster assignments, easy to determine centers
  - Now just a "weighted" center problem
- Given centers, easy to determine point → cluster assignments
  - Convert point ↔ center distances into similarities (e.g., $1/\text{dist}^{\alpha}$ or $e^{-(1/2)\text{dist}^2}$)
  - Set fractional membership to cluster j proportional to similarity to j's center.

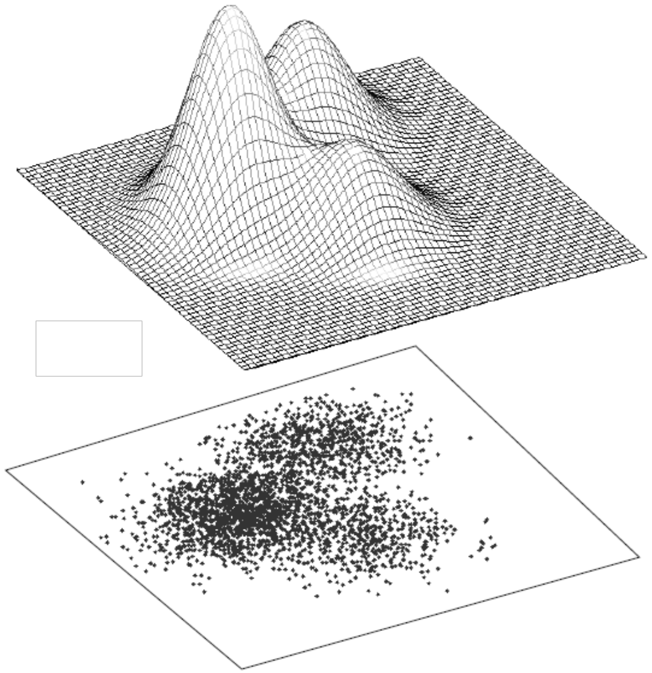


Gaussian Mixture Modeling

- Find parameters of a mixture of k Gaussian distributions most likely to have generated our data.

$$G(x \mid \mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left(-\frac{1}{2}(x-\mu)^\intercal \Sigma^{-1} (x-\mu)\right)$$

$$f(x) = \sum_{i=1}^{k} w_i G(x \mid \mu_i, \Sigma_i)$$



- Alternatively update:
  - – Explicit model parameters (w_i, μ_i, Σ_i)
  - – Latent parameters: p_{ij} = **Pr**[j^{th} point generated by i^{th} Gaussian]